Computational Voice Biomarkers for COPD: Cross-Lingual Validation on Danish Speech

Niklas Inderst
Technical University of Munich (TUM)
Chair of Health Informatics
niklas.inderst@tum.de

Clinical Applications Project - Supervisor: Andreas Triantafyllopoulos

***Abstract*—Chronic obstructive pulmonary disease (COPD) is a condition that affects respiration and can also affect speech production. This study tested whether Danish read speech contains enough acoustic information to distinguish speakers with COPD from controls in a binary classification setting.**

The study was not intended to create a clinical diagnostic tool but to test the idea in a Danish-speaking population. The study was based on a labeled cohort of 35 patients with COPD, and 43 controls, and explored two pipelines for feature extraction. The first pipeline used phrase segments from a fixed Danish reading passage, while the second used alignment-based vowel segments from the same reading passage. Acoustic features for both pipelines were extracted using eGeMAPSv02 with openSMILE and then fed into classical machine learning models, including logistic regression, SVM with RBF kernel, random forest, and gradient boosting.

The phrase-based pipeline produced the stronger result, reaching a speaker-level AUC of 0.715 and an accuracy of 0.653. The highest speaker-level AUC achieved by the passage-vowel pipeline was 0.669.

The results of the study suggest that there is sufficient acoustic information in Danish read speech to classify patients with COPD and healthy controls at the speaker level, but the overall effect under the present conditions is limited. The study found that using phrase segments from a fixed Danish reading passage gave stronger results than using vowel segments from the same passage. It also found that age was a significant factor in separating the groups in the cohort.

***Index Terms*—COPD, speech biomarkers, acoustic features, Danish speech, eGeMAPS, openSMILE, grouped cross-validation**

I. Introduction

A. Clinical motivation and problem

Chronic obstructive pulmonary disease, or COPD, is a chronic respiratory disorder characterized by persistent respiratory symptoms and airflow limitation [1]. As speaking is heavily dependent on breathing and air flow, it is reasonable to suspect that COPD might leave traces in speech. These traces would not necessarily be obvious to listeners. They might appear as minor differences in e.g. speaking rate, pausing behavior, loudness patterns, phonatory stability and other speech characteristics related to speaking under respiratory constraint. This is not to say that speech alone can be used to diagnose COPD, but rather that speech is a reasonable signal that may carry relevant information and is thus worth testing in a controlled manner.

The question posed in the study was more limited: Does standardized Danish speech contain sufficient signal for a machine-learning pipeline to distinguish between people with COPD and controls, better than chance? The aim was not to build a diagnostic product or make broad clinical claims. That is not what controlled data is for. It is there to test questions like this first. If no signal is found here, it is unlikely to appear in less controlled environments. And if it is found here, the next question is how stable and specific it is.

B. Why speech is worth testing

Speech is a practical signal because it is easy to collect and does not require invasive measurement, which is one reason it has attracted interest for remote and repeated monitoring in respiratory disease research [2], [3]. It can be recorded with simple hardware, repeated over time, and integrated into remote or digital workflows. This does not mean the signal will be automatically useful, but it explains the interest in speech-based signals as a topic of research in health-related machine learning. In cases where repeated contact with patients is difficult, the availability of an easy-to-collect signal could be useful. This is particularly true for chronic diseases, where changes over time may be as important or even more important than a single measurement.

Speech is influenced by many factors, such as age, sex, speaking style, fatigue, microphone quality, and task design. A model might also perform well because of factors that are not specific to the disease. This becomes even more important in the context of small clinical datasets. This study treated speech as a candidate signal, not as a solution in itself. It focused on whether a reproducible process could be created that identifies measurable structure in the data and on how strongly the results depended on the method of segmenting and evaluating the speech signal.

C. Research gap and why Danish matters

Other studies had already indicated that speech can carry information related to COPD. The most important conceptual background for our study is the paper by Mayr, Triantafyllopoulos, Batliner, Schuller, and Berghaus, which studies speech analysis in COPD patients during and after exacerbation [2]. It showed that there is clinically relevant variation in

COPD patients and that this variation can be approached using speech features and machine learning. However, there is also a difference between the task addressed in the original study and the task addressed in our study. The original study is about a within-COPD comparison, whereas our study is about binary COPD/non-COPD classification on Danish speech material.

This directly relates to the gap addressed in the paper. Even if the basic idea of using speech for COPD works for one particular dataset or speech material, it does not mean that it will automatically work for another. Speech analysis is very sensitive to different factors such as cohort structure, speech material, and recording conditions. A study that works for one type of speech material does not automatically mean that the same idea still works for other speech material and other task definitions. The fact that we used Danish material is relevant not because we think there is something special about the Danish language and COPD, but because it provides a different setting.

The current project addressed this gap through a controlled read-speech setup. The main analysis used Danish read speech and compared two different representations of the same material. In the first branch, the speech was segmented into phrase segments. In the second branch, vowel segments were derived from the same material using alignment-based processing. This way we were able to explore whether there was signal present in general, but also whether phrase segments or vowel segments derived from the material were more useful for speaker-level classification. This became one of the main methodological components of the project and allowed us to test whether finer segmentation was useful, or whether some of the more global properties of connected speech were more important.

D. Aim, research question, and contributions

Based on this setup, this paper aims to answer the following question: Is there sufficient signal in Danish read speech data to perform binary classification between COPD and non-COPD speakers? And how do phrase-level and passage-vowel-level pipelines compare under grouped speaker-level evaluation?

This paper makes four main contributions. First, it applies speech analysis using a reproducible pipeline to Danish COPD data in a binary classification setting. Second, it compares two segmentation pipelines based on the same read-speech data, one at the phrase level and one at the passage-vowel level. Third, it evaluates models using speaker-level metrics with grouped cross-validation and explicit aggregation over segment-level predictions, which reduces speaker leakage and makes the reported metrics better aligned with the actual prediction target. And finally, it investigates confounding effects, which were found to be significant.

II. Background and Related Work

A. COPD and likely effects on speech

COPD is a chronic disease affecting the lungs and is characterized by chronic respiratory symptoms and airflow limitation [1]. Dyspnea, cough, and sputum are among the usual signs of COPD in a patient, and it is often a progressive disease. From the perspective of speech production, COPD is a disease affecting the respiratory system, the body system that is responsible for speech production.

Speech is not produced by the larynx alone; it also depends heavily on the lungs. It is the coordinated combination of speech breathing, vocal fold vibration, and speech articulation. Breathing is not something peripheral to the process of speech, but one of the basic physical conditions for speech. Speech breathing research has long acknowledged that speech is composed of speech sounds and words, as well as of pauses and inhalations.

If there is a chronic respiratory disease affecting the lungs and thereby affecting speech breathing and/or speech intensity, there is a plausible basis for COPD affecting speech.

There are several different effects that COPD could cause in terms of speech. For example, COPD could plausibly lead to shorter breath groups, more frequent pausing, reduced loudness, and less regular temporal patterns. These effects are plausible, but they are also non-specific. Other factors and other conditions could produce similar patterns, so they are not enough on their own to determine COPD status in a patient.

The above physiological picture of COPD is also consistent with previous studies of COPD-related speech analysis. Previous studies have found a correlation between COPD and the speaking rate, inhalation, pauses, rhythm, as well as voice-quality-related features. For instance, Mayr et al. found that pulmonary function tests correlated with pause rate and rhythm variability in COPD patients, and that changes in symptom scores correlated with rhythm and stability of articulation [2]. Farrús et al. found that speech, especially prosody, was affected by physical effort and inhaled medication, and also stressed the importance of recording conditions in COPD-related speech analysis [3]. Taken together, these studies support the idea that COPD-related differences may be reflected in speech.

For this study, the question was whether the influence of COPD on respiration could be observed indirectly at the speaker level in a controlled Danish read-speech dataset

B. Speech biomarkers and engineered acoustic features

In this paper, a speech biomarker is defined as a measurable property of recorded speech that is associated with a health condition or health state. The focus is on whether the speech signal carries useful information. This can especially be relevant in small clinical datasets, in which performance above chance is not necessarily due to disease-specific clinical information.

A typical way of analyzing a speech signal is to compute a set of engineered acoustic features from the raw waveform and then use them to train classical machine learning models. This is a way of reducing a speech recording to a fixed representation that can be compared across speakers and utterances. In clinical and paralinguistic speech analysis, engineered features are still useful for two reasons. First, they are a standard way

of representing a speech recording. Second, a number of these features have a plausible connection to speech production, such as timing, spectral shape, loudness variation, pitch, and voice quality.

The features used in the current project were eGeMAPSv02, which is the more recent standardized variant of the Geneva Minimalistic Acoustic Parameter Set. The original GeMAPS publication was written with the goal of providing a more compact and theory-motivated baseline for automatic voice analysis [4]. Especially in settings where researchers were looking for something smaller and more interpretable than extremely large brute-force feature sets. It was explicitly intended as a baseline for paralinguistic and clinical speech analysis tasks, and the set is commonly used through openSMILE-based extraction workflows [4], [5]. This makes the choice appropriate for the current project as well.

The choice of the feature set is also sensible with regard to the scope of the current project. The objective of the current project was to test whether there was any signal present in the Danish data under a controlled and readable framework. This makes the use of a standardized engineered feature set appropriate, because it allows the process to be more easily inspected and the results more easily interpreted than would be the case with a black-box representation. This is why the current project used eGeMAPS and traditional models rather than starting directly with a more complex model.

C. Prior COPD speech work

Although the literature on speech analysis for COPD is still limited, its overall direction is quite consistent. Several studies have used speech and related vocal material as a potential signal for respiratory status in COPD, either by comparing COPD patients with healthy individuals or by comparing different states within COPD. While the studies differ from each other, those differences matter. Some studies used sustained vowels, while others used read speech; some used natural speech, while others focused on speech related to exacerbation.

Some studies focused on exacerbation detection or treatment-related change in COPD patients. Nallanthighal et al. used speech collected from COPD patients during stable and exacerbated states and compared three different approaches: acoustic features, low-level descriptive features with classification, and deep-learning-based speech-breathing models [6]. Later, Triantafyllopoulos et al. showed that read speech could distinguish pre- and post-treatment states in COPD patients, with strong effects from speaker normalization [7]. Mayr et al. followed the same general direction using a larger cohort and linked speech features to state classification as well as to CAT, BORG, and pulmonary function [2].

Another area of investigation has focused on more general speech differences associated with COPD. Farrús et al. investigated speech-based supervision of patient status in COPD patients and found that prosodic characteristics were particularly affected, while physical exertion, inhaled medication, and recording conditions also influenced the speech signal [3]. This is a useful study because it shows two things: It supports the idea that respiratory disease can affect speech, but also that recording conditions and task can affect the speech signal. This is part of the reason why it is difficult to compare performance metrics across COPD speech studies: the studies are simply too different.

The relevant takeaway from previous studies for this study is that there is enough prior evidence to justify a validation study. There is good reason to believe that COPD can affect speech, and previous studies have found measurable acoustic effects. However, there is still a great deal of fragmentation in the literature, and that creates room for a study whose primary contribution is not modeling novelty, but transfer to a new language and dataset.

D. Mayr et al. (2025): what they did and what differs here

The most relevant related-work paper for this project is Mayr et al. (2025) [2]. The paper analyzed speech in COPD patients during and after acute exacerbation. The data consisted of 50 COPD patients who were recorded during exacerbation and after treatment, before discharge. The modeling approach used spectral, prosodic, and temporal variability features, including eGeMAPS acoustic features, and an SVM classifier. The speech analysis achieved 84% accuracy in distinguishing between the two states and outperformed the baseline using only self-reported BORG and CAT scores. The paper also showed associations between speech features and clinical variables, including pulmonary function and pause rate/rhythm variability.

This paper is related to this project in three different ways: First, it shows that there is variation within COPD patients and that this variation is reflected in speech. Second, it is based on features and machine learning rather than end-to-end deep learning, which is more similar to this project. Third, it is well executed enough that it can be taken as a starting point rather than just inspiration.

However, there are significant differences between this paper and this project. The paper is about within-COPD state classification during and after exacerbation. The current project is about binary classification between COPD and non-COPD patients. This is a significant difference; a model trained to classify between states within COPD patients is not the same as one trained to classify between COPD and non-COPD patients. The current project is related to Mayr et al. (2025), but it doesn't study the same question.

There are, of course, other differences as well. The language setting is quite different, as this project is based on Danish read speech rather than the German-language clinical material of the Augsburg project. The segmentation strategy is also quite different. Rather than using a single representation of each recording, this project is based on comparing two representations of the same Danish read-speech material, one based on phrases and one based on passage-level vowels. This changes the question from "can speech reflect exacerbation state?" to "does Danish read speech contain enough speaker-level signal for binary COPD classification, and does this

signal vary depending on whether it is represented as phrases or as passage-level vowels?" This is a more circumscribed but still quite different question.

What carries over from the paper is mainly the idea that there are traces of respiratory disease in speech and that it is possible to design features and evaluation to detect those traces. What is not inherited is the task, language, population, or expected level of performance.

E. Positioning of this study

In that context, this paper is best viewed as a validation study. It does not put forward a new disease theory or a new clinical product, but rather, it takes an existing speech-based idea for COPD, relocates it to Danish read speech, and checks whether it still holds with a different label structure and a different segmentation design. Generalization, disease specificity, and other issues of this sort are not addressed.

III. Data and Task Definition

A. Cohort and labels

The supervised analysis was based on a dataset that consists of a speaker cohort of 78 individuals. At the speaker level, 43 were assigned to the control group while 35 were assigned to the COPD group. The cohort tended to be older and was heavily skewed towards female speakers. In the final label table, there were 65 female speakers and 13 male speakers. The age range of the speaker cohort was from 26 to 88 years, with a mean age of 67.49 years and a median age of 71.5 years. These numbers already show that age cannot be considered a minor background factor in the analysis, and has to be kept in mind from the beginning.

The label used in this paper was binary. Each speaker was given a binary label of either COPD or control. It was not a matter of predicting symptom severity, disease stage, or whether the patient was in an exacerbation. It was also not a longitudinal problem, but rather a matter of whether the available Danish speech would be sufficient to differentiate these two groups of speakers better than chance.

It should also be noted that the binary nature of the problem in this paper should be considered carefully. The use of a COPD or control label does not mean that any speech differences found between these two groups have to be related only to the presence or absence of COPD itself. In a population of this size, a variety of other factors could also contribute to the separation between the two groups. Age would clearly be one of the main factors in the population used in this paper. Later analysis showed that age likely played a significant role in this cohort.

B. Recording material and tasks

The broader dataset contained more than one form of recording. At the repository level, there were three types of tasks: a read passage, a series of sustained vowels, and cough recordings. However, the main paper focused on the read speech material. This was a deliberate choice, as the read passage gave a common linguistic form across speakers and allowed the comparison of two segmentation methods on exactly the same underlying speech content.

TABLE I
Cohort overview.

Variable	Value
Labeled speakers	78
Control speakers	43
COPD speakers	35
Female speakers	65
Male speakers	13
Mean age	67.49
Median age	71.5
Age range	26–88

TABLE II
Recording and task overview.

Recording type	Main analysis	Role in paper
Read passage	Yes	Main source material for the phrase and passage-vowel pipelines
Sustained vowels	No	Secondary repository branch and additional comparison
Cough recordings	No	Present in the dataset, but not used in this paper

The first form of read speech was based on phrase segments. The Danish read passage was segmented into fixed phrase units, which retained short stretches of connected speech and therefore preserved information about timing, local prosody, and speech over a larger respiratory window than a phoneme or vowel. The second form of read speech was based on vowel segments extracted from the same Danish passage. This yielded a much finer set of speech segments and therefore a more local speech event, with many more segments per speaker.

A third form of speech in the repository was based on sustained vowels. While not used in the main part of the work, it was useful as a comparison and eventually gave interesting results.

C. Final task definition and label setup

The final supervised task was the classification of a speaker: given speech from a certain speaker, it was to be predicted whether the speaker belonged to the COPD group or the control group. However, the models were not trained on a single vector per speaker from the start. Instead, the speech was segmented, and acoustic features were extracted from these segments. In the phrase branch, these segments were phrases, and in the second branch, these segments were vowel segments from the same passage, as already discussed.

The observations were at the segment level, but the prediction target was at the speaker level. This was a crucial distinction in the entire experiment. The paper did not treat every phrase and every vowel segment as if it were an independent person-level sample. This would have overstated and distorted the evidence in the data. Many segments could come from the same person, but they would still represent the same label, the same recording context, and the same person.

Therefore, the predictions at the segment level were later aggregated to the speaker level for evaluation. The question here was less whether a single isolated phrase or a single isolated vowel segment could be correctly classified in a vacuum, but rather whether the entire segment-level evidence available for a speaker could support a classification decision at the speaker level.

D. Read-speech passage and segment inventory

The major speech input came from a predefined Danish reading passage. In the phrase pipeline, this was segmented into 20 phrase segments per speaker. The raw phrase inventory had 1,620 segments from 81 speaker IDs. The supervised phrase feature table had 1,560 segments from 78 labeled speakers after merging with the final label table and removing unlabeled speakers. Each speaker had an equal number of phrase segments.

The passage-vowel branch started from the same Danish reading passage. In this branch, alignment-based segmentation was performed to obtain vowel segments. The raw inventory had a significantly larger number of segments compared to the phrase branch. The raw vowel segmentation table had 17,605 vowel segments from 81 speaker IDs. The final vowel feature table had 12,327 vowel segments from 78 labeled speakers after merging with the final label table. The number of vowel segments was not equal for each speaker, and the actual number of segments depended on the details of the segmentation.

E. Data caveats before modeling

There were some data characteristics that were important enough that they had to be mentioned before moving on to the methods section. The first was the cohort size. With 78 speakers, the dataset was suitable for a pilot-style evaluation, but too small to support strong clinical machine-learning claims. It was therefore necessary to be very careful with any results that came from this data.

The second was that of cohort composition. This was an older and gender-skewed speaker population, and that was going to become very important in terms of interpretation of the results, as in some cases, it was possible that the machine learning algorithm could separate speakers based on demographic characteristics, not necessarily on disease-based speech characteristics. This was a real issue in the dataset and had to be considered throughout the paper.

The third was that of inventory and supervised cohorts. There were 81 segmented speaker IDs, but only 78 labeled speakers. This was not an issue in and of itself, but it was an indication that supervised and processing stages were not necessarily conducted on exactly the same data.

The fourth was that of units. There were far more segments per speaker in the passage-vowel branch than in the phrase branch. Not because there were more speakers, but because there were more segments per speaker. In fact, this was one reason why speaker-level evaluation was necessary.

Overall, these issues did not invalidate the results, but they did define how those results should be viewed and interpreted. This was a study asking a focused question on a small, structured Danish cohort.

IV. Methods

A. Overall pipeline overview

The entire pipeline followed a similar pattern in both of the major branches of analysis. First, speaker labels and demographic metadata were gathered at the speaker level. Second, the Danish read-speech recordings were segmented into smaller analysis units. This gave a phrase-based and a passage-vowel-based view of the speech material. Third, acoustic features were extracted from each segment using the same family of standard acoustic features. Fourth, classical machine-learning models were trained using the segment-based tables of acoustic features. Fifth, grouped cross-validation was used at the speaker level during evaluation. This ensured that no speaker was used in both the training and test sets at the same time. Finally, segment-based model results were combined back to the speaker level and compared.

The entire process was designed in a way that tried to keep the focus on the segmentation of the speech material rather than on the changes in the data, features, or the evaluation process itself.

A third, sustained-vowel branch was also present in the repository and was used as a secondary analysis. It is discussed in more detail later in this Method section. It was not the main methodological point of interest in the research though. Fig. 1 summarizes this shared workflow and shows where the two main read-speech branches diverge.

B. Phrase segmentation pipeline

The first main branch used phrase-level segments derived from a fixed Danish read passage. Each speaker read the same text, which made it possible to compare speakers on shared linguistic material rather than on different spontaneous utterances. Uncontrolled lexical and structural variation can blur acoustic comparisons, especially in a small dataset. A standardized passage does not remove all variability, but it reduces one major source of it.

To build the phrase pipeline, the read-passage audio was aligned with the transcript using a forced-alignment workflow. The output of this step was stored in TextGrid form and then converted into a structured phrase table [8]. The alignment stage was necessary because the project did not rely on one single pre-cut recording per phrase. It started with full read-passage audio and then used the alignment output to determine segment boundaries in a way that was reproducible.

The phrase table contained 20 phrase intervals per speaker. These intervals corresponded to fixed parts of the Danish passage and were indexed accordingly across speakers. In the raw phrase inventory, this yielded 1,620 phrase rows across 81 speaker IDs. After merging with the final supervised label table and excluding unlabeled cases, the feature table used for modelling contained 1,560 phrase segments from 78 labeled

TABLE III
MAIN READ-SPEECH SEGMENT INVENTORIES.

Table	Rows	Speakers	Role
phrases.csv	1,620	81	Raw phrase segmentation inventory
phrase_features_with_labels.csv	1,560	78	Final supervised phrase table
vowels.csv	17,605	81	Raw passage-vowel segmentation inventory
vowel_features_with_labels.csv	12,327	78	Final supervised vowel table

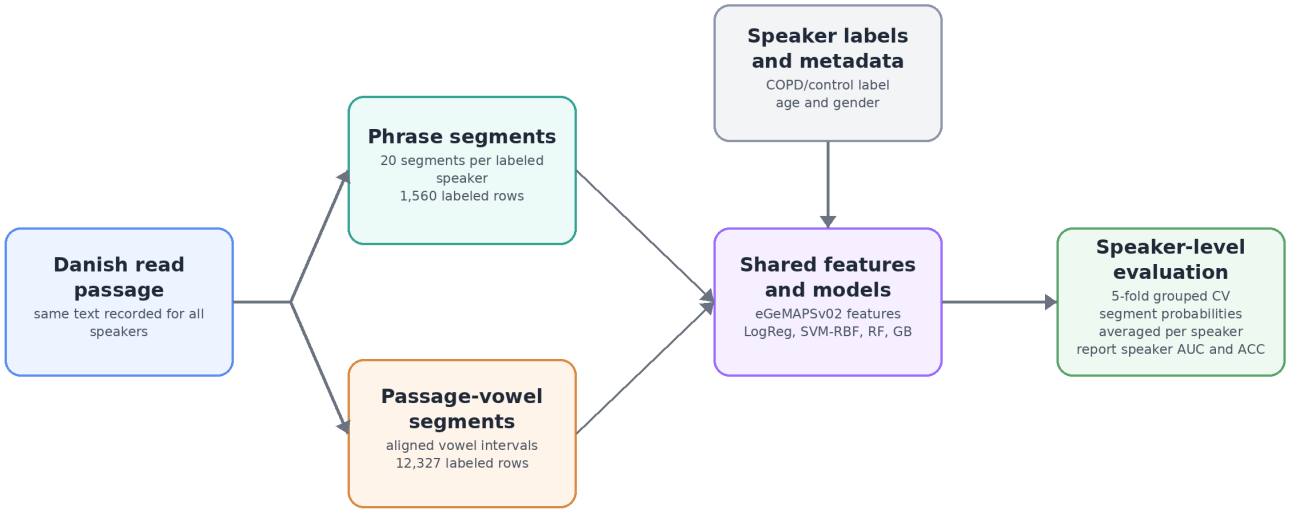

Fig. 1. Overall pipeline schematic. End-to-end flow used in the main read-speech experiment. The two main branches differ only in segmentation unit; labels, features, models, and evaluation are shared.

speakers. Every labeled speaker therefore contributed the same number of phrase units to this branch.

A regular segment structure is one advantage of the phrase branch, as it gives a balanced segment structure across speakers and preserves more context than a very short phonetic unit would. A phrase still contains timing patterns, local pause behavior, prosodic movement, and short-range respiratory effects that may be weakened or lost when the signal is broken into much smaller pieces.

Once phrase boundaries had been defined, each phrase segment was treated as one acoustic unit for feature extraction. The subsequent modelling step therefore operated on a segment-level table in which each row represented one phrase from one speaker, together with its acoustic features and speaker-level label. The label itself remained attached at the speaker level, but the observations used for model fitting were phrase-level rows.

This branch was designed to test whether breaking the same standardized passage into short but still meaningful chunks of connected speech preserves enough information for speaker-level COPD classification. Fig. 2 shows this phrase-based branch in schematic form.

C. Passage-vowel pipeline

The second main branch used the same Danish read passage but represented it at a much finer level. Instead of cutting the passage into phrase units, the alignment output was used to isolate vowel intervals from the passage. This produced a passage-vowel inventory rather than a phrase inventory. In the phrase branch, each unit still contains a short amount of running speech. In the vowel branch, these units are just much shorter and more local.

The passage-vowel pipeline used the same transcript-aligned read-speech material as the phrase pipeline. Again, a similar alignment process was used to obtain interval information from which vowel segments could be extracted [8]. These vowel segments were collected in a vowel table with segment identity, timing information, and speaker ID. In the raw inventory, the vowel table contained 17,605 rows across 81 speaker IDs.

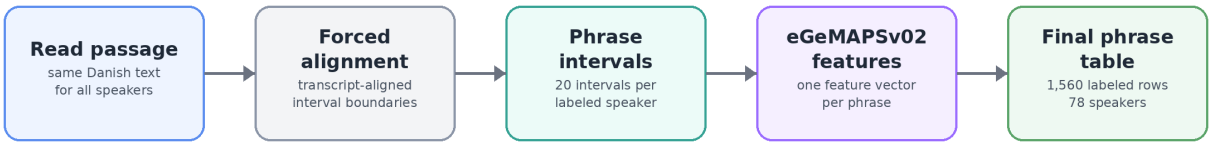

Fig. 2. Phrase pipeline. The read passage was aligned to the transcript, converted into phrase intervals, and represented with one eGeMAPSv02 feature vector per phrase segment.

After the label merge and filtering to the supervised cohort, the final modeling table contained 12,327 vowel segments from 78 labeled speakers.

Unlike the phrase branch, the number of segments per speaker was not fixed. The final number of retained vowel intervals depended on the alignment output and the amount of clean extractable material from the passage for a speaker. Among the labeled speakers, the number of vowel segments varied a lot.

This branch was worth testing, however, as very short vowel segments may contain detailed phonatory and spectral properties that are lost when averaging over a longer stretch of running speech. If COPD has an impact on voicing stability, spectral properties, and other local acoustic properties, a finer segmentation may in principle make these properties easier to detect. On the other hand, this representation also throws away a significant amount of information about the larger context, so the vowel branch is another trade-off between contextual and local information.

Just like in the phrase branch, each segment was again represented by a row in the feature table. These rows were used for training, but the prediction target was still the speaker. So the main purpose of the vowel pipeline was to test whether a large number of fine-grained vowel properties, combined at the speaker level, would improve classification over the representation based on phrases. Fig. 3 shows the passage-vowel branch and its finer segmentation logic.

D. Feature extraction with eGeMAPSv02

Acoustic features were extracted from all analysis segments using **eGeMAPSv02**. This was the main feature representation across the project and was used in both read-speech branches. The same general feature family was applied across branches to keep the comparison fair. If different acoustic representations had been used for phrases and vowels, any later performance difference would have been harder to interpret.

The choice of eGeMAPSv02 fit the scope of the study well, as it is a standardized and widely used engineered feature set for paralinguistic speech analysis. A standardized feature set makes it easier to describe the method used and to compare its behavior across different segmentation strategies.

Feature extraction was carried out with **openSMILE**, using the eGeMAPSv02 functionals configuration [4], [5]. The resulting representation consisted of **88 acoustic features** per segment. These features describe several high-level families that are commonly used in clinical or paralinguistic speech analysis. They included pitch-related descriptors, loudness-related descriptors, spectral and cepstral summaries, voice-quality and perturbation measures such as jitter- and shimmer-related quantities, and additional summaries describing harmonicity and formant-related structure.

The three main processed feature tables followed the same general logic. Each row corresponded to one segment, each segment carried a speaker ID, and each row combined acoustic features with speaker-level metadata and labels. In the phrase branch, the final labelled feature table contained 1,560 rows and 88 acoustic features per row, together with phrase metadata and speaker-level variables. In the passage-vowel branch, the final labelled table contained 12,327 rows with the same acoustic feature dimension.

Using the same feature family across branches did not make the two representations identical. Phrases and vowels differ in duration, local structure, and what parts of the speech signal they preserve. But it did remove one unnecessary source of variation. Any later difference between the branches could therefore be interpreted more directly as a difference in the unit of analysis, not as a difference in the basic acoustic feature extractor.

E. Model families

The supervised comparison was performed with four classical machine-learning model families:

Logistic Regression
Support Vector Machine with RBF kernel
Random Forest
Gradient Boosting

This was an appropriate selection of models given the data at hand. The data was small at the speaker level, and the feature space was tabular. Moreover, the primary concern was comparing segmentation strategies, which was in this context more important than aggressive model optimization. Given these circumstances, classical models were more appropriate than large end-to-end models applied directly to waveform data.

The models were implemented in scikit-learn [9]. For Logistic Regression and the RBF SVM, a standardization step was added, since both models are sensitive to feature scaling. For Logistic Regression, a maximum of 2000 iterations was specified. For the SVM, probability output, balanced class weights, C = 1.0, and gamma = “scale” were used. For RF, 400 estimators, balanced class weights, and a fixed random state were specified. For GB, 350 estimators, a learning rate of 0.05, and max_depth = 3 were specified. These models were specified in this manner because the project was not particularly focused on hyperparameter search, and since the data was small, heavy tuning was not particularly sensible here.

Fig. 3. Passage-vowel pipeline. The same read passage was segmented more finely by extracting vowel intervals from the aligned transcript, which produced many more local segments per speaker.

The reason for choosing these models was mainly that they provide a useful spread. Logistic Regression provides a linear model. The RBF SVM provides a nonlinear model in the same feature space. RF and GB provide tree-based models with different ensemble behavior. This made it possible to see whether any observed signal was constant across models or depended strongly on one model or modeling family.

F. Validation design and speaker-level aggregation

The design of the evaluation was perhaps one of the most significant aspects of the project. The reason for this is mainly that the modeling tables were segment-based, and a poor split strategy would have led to highly inflated results. A speaker in the data provided a large number of rows in the feature table, 20 rows in the phrase branch and a much larger number in the passage-vowel branch. If a poor split strategy was used, with no consideration for the speaker, it is very likely that the same speaker would be in both the training and test data. This would be a poor test of speaker-level generalization, and would be considered a leakage. Therefore, a 5-fold StratifiedGroupKFold was used with shuffle=True and random_state=42. This ensured that all the data from a single speaker stayed in a single fold. Stratification was also used to ensure that the composition of the COPD/control classes remained at least relatively stable. This is important especially with such a small dataset of only 78 labeled speakers. A simple grouped split would have led to a highly unbalanced composition in some folds.

The evaluation was performed in two stages.

In the first stage of the process, the fitting of the model happened at the segment level. For a single training fold, the model was presented with all the segment rows from the speakers in that fold. The acoustic feature matrix was created from the segment table. Depending on the setup of the analysis, age and gender were either included in the matrix or left out. The model then generated predicted probabilities for the held-out segments. At this point in the process, the prediction unit remained the segment.

The second stage of the process involved aggregating the segment-level results to the speaker level. All the predicted probabilities from the same held-out speaker were averaged. The result was a single average probability per speaker. A class label was then created for the speaker by thresholding the average probability at 0.5. The same probabilities were used to calculate the AUC at the speaker level.

This design matched the actual question of the paper. The study was not trying to classify isolated phrases or isolated vowels as if they were independent clinical cases but rather trying to classify speakers. Segment-level modelling was only the internal mechanism that allowed the project to use multiple observations per speaker. The final target remained the individual person.

The main metrics were speaker-level AUC and speaker-level accuracy. Speaker-level AUC was calculated from the averaged probabilities. Speaker-level accuracy was calculated after the probabilities were converted into a binary prediction by thresholding at 0.5. Segment-level metrics were also available in the evaluation. However, they were not the main metrics.

This type of evaluation was especially necessary for the passage-vowel branch, as it yielded many more rows than the phrase branch. Without grouped evaluation, the number of samples would have appeared much larger than the actual number of people in the study. Grouped speaker-level cross-validation was necessary here to correct that and keep the comparison between the branches fair. A branch with more segments should not be considered better just because it has more segments, thereby providing more opportunities for leakage.

The same idea also matter for how the results should be interpreted. If a branch performed better, it did not mean that all of the individual segments were very informative. It meant that, after combining all the evidence from all the held-out segments of a speaker, the model was better able to discriminate between speakers with COPD and those without it.

Another benefit of this setup is that it kept the results close to the real scale of the data. With only 78 labeled speakers, no evaluation strategy could make the study larger than it was. So the grouped strategy did not solve this sample-size problem, but instead prevented the segment structure from hiding it and thus keeping the results close to the real scale of the data. Fig. 4 illustrates this speaker-level evaluation logic.

G. Age/gender handling and ablation design

Age and gender were available in the label table and were treated as optional input variables rather than as necessary background information. This matters in a dataset like this, because age and gender could be important factors in classification. The paper had to consider two different questions: How much signal could be derived from speech features? And how much of that signal could be due to demographic structure?

In the preprocessing step, age was converted into numerical form. Gender was normalized and then encoded as a binary variable. The modeling code had two different versions of the feature matrix. In one case, age and gender were concatenated

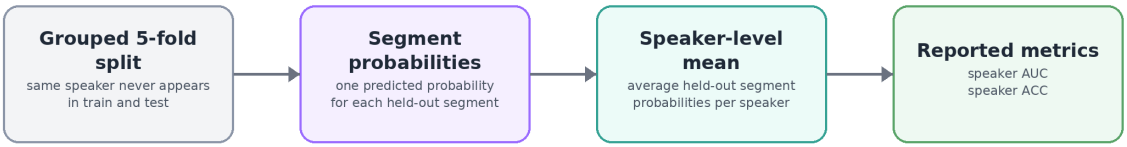

Fig. 4. Speaker-level evaluation logic. Segment rows are used for training, but splitting and reporting stay at the speaker level.

to the acoustic feature vector. In the other case, only acoustic features were used.

The ablation was not meant to argue that age and gender should never be used. In some applied settings, demographic information may be available and may improve performance. The goal was to test whether classification depended in part on demographic structure that could make the task easier without necessarily depending on disease-specific speech properties. This question is important for this paper as well because this is a population of older speakers whose age distribution could overlap with COPD.

The ablation test is part of the interpretation of results as well as part of modeling. If the classification results degrade when age and gender are removed, that is not necessarily a problem with the speech features. However, it does affect what it means to claim a particular level of performance.

H. Secondary sustained-vowel branch

In addition to the two main branches for read speech, the repository contained a third branch for sustained vowels. This branch used separately recorded vowel sounds and not vowel intervals derived from the passage. From a methodological standpoint, this makes it a different task. Intervals derived from the passage are sounds produced in the context of running speech. In contrast, sustained vowels are isolated phonations produced outside the context of the passage.

The sustained-vowel inventory was created from separate vowel sounds, each stored at the speaker level. After indexing and merging with the labels, the labeled sustained-vowel feature table contained 366 rows from 74 labeled speakers. As in the previous two cases, each row contained eGeMAPSv02 feature data, speaker information, and the binary COPD label. As with the previous two branches, the same modeling logic was applied to this branch, which includes grouped speaker-level evaluation.

The sustained-vowel branch was a secondary branch for a number of reasons. First, the question under consideration was related to the Danish passage and the two methods for segmenting it. Second, sustained vowels simply do not answer the same question, as they remove a significant amount of the word and running-speech context and focus on a simpler vocal task. However, they are simply additional context and do not replace the original question.

The sustained-vowel branch was still worth evaluating. While it does not directly answer the original question, it does offer another way to look at the same general classification problem. This made it possible to test whether the classification signal was specific to read speech or whether it was also present in more controlled vowel recordings. This branch is reported later in the results section.

V. Results

A. Dataset and QC results

The supervised cohort consisted of 78 labeled speakers, with 43 controls and 35 with COPD. The processed read-speech tables, which were used for the main experiment, contained 1,560 labeled phrase segments and 12,327 labeled passage-vowel segments.

The QC figures confirm the shape of the data used in the main experiment. This experiment was small in terms of speaker count, although the segment tables were considerable in size, which is particularly significant for the passage-vowel branch, where a considerable number of rows came from the same speaker and thus were not independent cases. This is another reason speaker-grouped evaluation was necessary. Fig. 5 gives a compact overview of the labeled cohort and the two main read-speech tables used in the analysis.

B. Main phrase results

The phrase branch produced the stronger of the two main read-speech results. Of all four classical models, Logistic

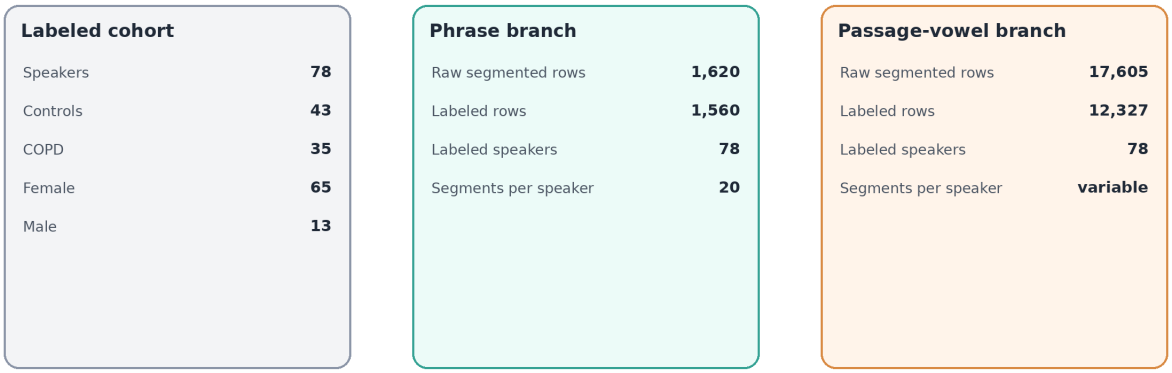

Fig. 5. Dataset QC overview. Compact summary of the labeled cohort and the two main read-speech tables used in the analysis.

TABLE IV
PHRASE BRANCH RESULTS.

Model	Speaker AUC	Speaker ACC	Segment AUC	Segment ACC
Logistic Regression	0.715	0.653	0.662	0.610
Random Forest	0.690	0.602	0.636	0.592
SVM-RBF	0.684	0.615	0.642	0.595
Gradient Boosting	0.544	0.552	0.540	0.535

TABLE V
PASSAGE-VOWEL BRANCH RESULTS.

Model	Speaker AUC	Speaker ACC	Segment AUC	Segment ACC
SVM-RBF	0.669	0.637	0.619	0.578
Gradient Boosting	0.667	0.640	0.652	0.643
Logistic Regression	0.658	0.603	0.614	0.581
Random Forest	0.650	0.677	0.629	0.593

Regression achieved the best result on the main speaker-level metric: a mean speaker AUC of 0.715 and a mean speaker accuracy of 0.653. Random Forest came second with a speaker AUC of 0.690, then SVM-RBF with a speaker AUC of 0.684, while Gradient Boosting was much weaker in this branch with a speaker AUC of 0.544. The same order was maintained on the second metric: speaker accuracy, where Logistic Regression again outperformed all other classical models.

The COPD and control speakers were separated better than chance in the phrase branch. However, the overall performance was only moderate. A speaker AUC of about 0.72 is meaningful in a small grouped evaluation. Nevertheless, it is still far from being strong enough to make any diagnostic claim. The result is much narrower in scope than any diagnostic claim. It mainly indicates that there is some signal present in the data at the speaker level.

Fold-level behavior also varied noticeably. This is naturally not surprising in the case of the grouped 5-fold setting with 78 speakers. However, it does have implications for how the result should be interpreted. There were folds that were better than others, especially for the weaker models. The phrase "result" is thus better interpreted as meaning "moderate performance with non-trivial fold-to-fold variation."

C. Main passage-vowel results

The passage-vowel branch also resulted in above-chance speaker-level classification, though it did not match the phrase-branch results on the main metric. Again, the best speaker AUC was achieved by SVM-RBF at 0.669. Gradient Boosting was a close second at 0.667, followed by Logistic Regression at 0.658 and Random Forest at 0.650. If speaker accuracy is taken as a separate measure, Random Forest again scored the highest at 0.677, followed by Gradient Boosting at 0.640, SVM-RBF at 0.637, and Logistic Regression at 0.603.

The results of the passage-vowel branch differed from those of the phrase branch. In the passage-vowel branch, SVM-RBF gave the best speaker AUC, while Random Forest gave the best speaker accuracy. No particular model family scored consistently highest in the passage-vowel results. However, the main finding remained the same: speaker-level classification using passage-vowel segments worked, but not as well as the phrase-based results.

The passage-vowel results were not poor, but they were not strong enough to show an advantage for finer segmentation. While the passage-vowel branch generated a much larger number of segment rows than the phrase branch, this finer segmentation did not lead to stronger speaker-level classification.

The fold tables again showed visible variation across the grouped splits, especially for Random Forest and Gradient Boosting. This is another reason not to read too much into the result. The branch was informative, but not stable enough to support stronger claims than that.

D. Direct comparison of phrase and passage-vowel pipelines

In terms of the primary metric, speaker-level AUC, the phrase pipeline was superior to the passage-vowel pipeline. For speaker-level AUC, the best result in the phrase pipeline was 0.715, achieved by Logistic Regression. For the passage-vowel pipeline, the best result was 0.669, achieved by SVM-RBF. So, there was a difference of about 0.046 in speaker-level AUC. For speaker-level accuracy, passage-vowel Random Forest achieved 0.677, thus outperforming the best result in the phrase pipeline, 0.653. Since speaker-level AUC was taken

TABLE VI
PHRASE AGE/GENDER ABLATION (RANDOM FOREST).

Setting	Speaker AUC	Speaker ACC	AUC change	ACC change
Acoustic only	0.664	0.552	–	–
With age/gender	0.704	0.614	+0.040	+0.062

as the primary comparison criterion, the phrase branch was superior.

The comparison was valid since both branches were exposed to the same data, the same acoustic feature family, and the same model set. The only difference between the two was that they used two different segmentation levels. Under these conditions, the phrase branch performed better on the primary metric. This is because, in this data, preserving connected speech in phrases was more useful than preserving information in aligned vowels. This is further explained in Section 6.2.

The comparison also shows that, in spite of generating so many segment rows, the passage-vowel branch was not able to achieve better speaker-level discrimination. This suggests that the number of segments alone did not determine performance; what mattered was the type of information preserved in the analysis units. Fig. 6 summarizes the main branch comparison on the paper's primary metric.

E. Age/gender ablation

The age/gender ablation gave one of the clearest confounder-related results in the repository. For the phrase branch, a stored Random Forest comparison showed that adding age and gender improved speaker AUC from 0.664 to 0.704 and speaker accuracy from 0.552 to 0.614. The absolute gains were 0.040 in speaker AUC and 0.062 in speaker accuracy. The effect is substantial. It shows that demographic information contributed materially to performance in at least this part of the analysis.

The result from the acoustic-only model still remained above chance. However, the gain from adding age and gender suggests that some part of the observed separation likely came from demographic structure rather than speech-derived disease information alone. In this dataset, age is certainly not a minor factor.

F. Secondary sustained-vowel results

The secondary sustained-vowel branch had higher speaker-level AUC results than either of the two main read-speech branches. With age and gender included, the best stored result was Random Forest with a speaker AUC of 0.793 and a speaker accuracy of 0.730. Without age and gender, the best speaker AUC was 0.801 with Gradient Boosting, with Random Forest very close at 0.800.

The sustained-vowel result provides context, but it is a secondary result because it comes from a different speech task. It suggests that more controlled phonation may have a clearer signal in the repository than read speech does. It does not replace the main result of the paper, which compares phrase versus passage-vowel results on Danish read speech.

TABLE VII
SECONDARY SUSTAINED-VOWEL RESULTS.

Branch	Best model	Speaker AUC	Speaker ACC
With age/gender	Random Forest	0.793	0.730
Without age/gender	Gradient Boosting	0.801	0.663

VI. DISCUSSION

A. What the main result actually shows

The main result was that the read-speech data contained a measurable but only moderate classification signal. The best phrase-level pipeline achieved a speaker AUC of about 0.72, while the best passage-vowel pipeline achieved an AUC of about 0.67. These are not trivial results, especially considering that the study included only 78 speakers and used strict fold separation by speaker. At the same time, they are not strong results.

The results need to be read carefully. It should not be treated as evidence that speech can already serve as a reliable diagnostic tool for COPD. It should be read more narrowly. In this dataset, under a controlled setup, speech carried enough information for above-chance classification at the speaker level. That supports the basic idea behind the project. It does not justify stronger claims than that.

It is worth noting that the evaluation design also influences how the result should be interpreted. The fact that the models were evaluated with grouped cross-validation and speaker-level aggregation means that the results are closer to the true scale of the problem than they would have been under segment-level evaluation alone. The phrase and vowel tables had many more rows than speakers, especially in the vowel branch. This means that without speaker grouping, these extra rows could easily have led to overestimation. The fact that moderate performance was still present under grouped evaluation makes the result more credible, since it suggests that the signal was not dependent on row counts or leakage.

The Danish data contained useful information, and that signal remained visible under a careful evaluation design. This is worth reporting, although it does not mean that the method could be used in practice.

In addition, the result provides some limited support for the original project idea, namely that COPD-related respiratory differences could influence speech production in a way that could be detected acoustically. While it does not show that all of the signal is disease-specific, it does show that the idea remains plausible in a different language and under a different binary classification task.

B. Why phrases may outperform passage-vowels

The phrase-versus-vowel comparison gives the clearest technical result in the paper. On the main metric, speaker AUC, phrase-level modelling worked better than passage-vowel modelling. A likely reason is that phrases preserve more of the structure that COPD could affect.

Fig. 6. Main results comparison. Best speaker-level performance in the two main read-speech branches. The primary comparison metric in the paper is speaker AUC.

A phrase is long enough to retain local prosody, pausing behavior, rhythm, phrase duration, and breath management across several words. Those properties are closely tied to respiratory support and timing. If COPD affects speech through breath control, effort, or pacing, then some of that information may only become visible over a larger window than a single vowel can provide. Phrase segments keep part of that window intact.

Passage-derived vowels capture something else. They isolate shorter acoustic events and may reflect local voice quality or phonatory behavior more directly. That can be useful. But the cost is loss of context. Once the passage is cut into many individual vowels, much of the longer-range timing structure disappears. The model then sees more units, but each unit carries less information about how speech unfolds over time. That may be one reason why the vowel branch produced many more rows without producing a better speaker AUC.

There is also a practical issue. Passage-vowel extraction depends more heavily on alignment quality. Small segmentation errors matter more when the unit itself is short. A phrase boundary that is slightly imperfect may still leave most of the relevant acoustic material intact. A vowel boundary that is slightly off can change the segment much more. That does not mean the vowel pipeline was invalid. It means it was probably more sensitive to local noise.

This interpretation should still stay modest. The project did not run a controlled causal test of why phrases performed better. It only showed that they did. The most reasonable reading is that phrase units preserved useful connected-speech information that the vowel representation partly lost.

C. What the sustained-vowel branch changes and does not change

The sustained-vowel branch adds useful context, but it does not overturn the main result of the paper. In the repository, that branch reached higher speaker-level AUC values than either of the two main read-speech branches. The best stored result with age and gender reached a speaker AUC of 0.793, and the best result without age and gender reached 0.801.

This matters because it suggests that a more controlled phonation task may preserve a clearer signal than running read speech in this repository. Sustained vowels reduce variation from lexical content, speaking rate across phrases, and passage-level timing differences. That can make local phonatory and spectral effects easier to isolate.

At the same time, the sustained-vowel branch comes from a different speech task and a different inventory structure. It therefore cannot replace the main comparison in this paper, which was designed around Danish read speech and the contrast between phrase-level and passage-vowel segmentation of the same passage material. The sustained-vowel result is best read as supporting context, not as the headline result.

It also makes the interpretation of the project slightly more nuanced. The repository does not point to one universally best speech unit for every setup. Instead, it suggests that the most useful unit depends on the task. Within the two read-speech branches studied here, phrases worked better than passage-derived vowels on the primary metric. Across the wider repository, however, a separate sustained-vowel task produced the highest stored AUC.

D. Confounding, especially age

Age is another important factor in this project. The cohort was older overall. The binary label was learned from speakers rather than from disease status in isolation. This means that a variable correlated with group membership can contribute to classification. Age is an obvious candidate. Age can affect speech in many ways even in the absence of COPD. Breathing efficiency, phonatory stability, speaking rate, loudness control, and overall voice characteristics can all vary with age. The model can pick up some of that structure whether or not it reflects COPD-specific effects. The stored ablation result

supports this. In the phrase branch, adding age and gender increased speaker AUC from 0.664 to 0.704 and accuracy from 0.552 to 0.614. The acoustic-only system still performed better than chance, but some of the observed separation was likely due to demographic information.

This changes the interpretation of the results. If some of the predictive ability is coming from age or gender, then we are not seeing a pure disease marker. We are seeing a marker that is both acoustic and cohort related.

However, knowing what the confound is not a failing of the project. It is one of the strengths of the project. Small speech datasets can have results that look cleaner than they really are because confounding is not accounted for. Here, at least, we made this confound visible.

The sustained-vowel branch adds one more degree of complexity to the results. In that branch, we were not seeing a pure "with age and gender is always better" pattern. The best stored speaker AUC for that branch was actually from a no-age/gender setup. So confounding effects were present, but they were not necessarily the same across all branches. That is one reason to be careful. The issue is that age probably explains some of the effect we are seeing, and the size of that effect may vary depending on the task or the model.

E. Relation to the original paper and cross-lingual transfer

The findings are certainly in line with the broader idea of the original Mayr et al. paper. This is the sense in which the current study can be said to follow the original. However, there are important qualifications to be made. The original study dealt with a different problem. Specifically, the original study dealt with the problem of classifying states within COPD during versus after an exacerbation. This study dealt with the problem of classifying COPD vs. non-COPD. These are not the same problem. The label set, the sources of information, and the sources of noise are all different. Therefore, the current study should not be considered a simple re-run of the original.

The previous work inspired the notion that respiratory disease impacts speech in a measurable way. This work tested whether the basic notion still held when applied to a Danish dataset with a different classification target. The answer is yes, but only to a limited extent. There was measurable signal. However, the work also demonstrated that transfer cannot simply be assumed to work well. The results were moderate, and some of the signal may be due to demographic factors. So, the idea of cross-lingual transfer was plausible, but limited.

F. Practical value

Despite the limitations, speech remains worth studying. It is easy to collect, non-invasive, and naturally fits into digital processes [2], [3]. This work also demonstrated that a clear and transparent methodology can be developed around the basic idea using common acoustic features, grouped evaluation, and speaker-level aggregation.

But the current results do not warrant any claim for clinical use. There is insufficient evidence for diagnostic replacement, and insufficient evidence for a screening tool without stronger validation. The contribution of the current work is more specific: a careful evaluation on a Danish dataset, a direct comparison of phrase-level and passage-derived vowel representations, and a realistic assessment of the signal and its limitations.

VII. Limitations

A. Small cohort and uneven group structure

The main limitations are the size and composition of the dataset. The dataset had 78 labeled speakers, with 35 speakers in the COPD group and 43 speakers in the control group. That is sufficient for a first test but is not a large data set for a speaker-level classification task. The performance can vary depending on the data splits for the folds, which speakers are included in each split, and what model is used. The reduction of leakage is possible with grouped cross-validation, but it does not eliminate the inherent uncertainty due to the data set size.

The composition of the data set is also important. The data set was not well balanced for male and female speakers. The data set had 65 female speakers and only 13 male speakers. That makes it challenging to determine whether the learned signal is robust across groups. Additionally, it is challenging to determine how well the results can be generalized beyond the composition of the data set. The age composition also plays a role since the data set was dominated by older people.

B. Confounding by age and demographic variables

Age was a major limitation in interpreting the results. The study demonstrated that demographic features have a bearing on performance, particularly for the phrase branch. When age and gender features were incorporated into the experiment, speaker-level AUC increased compared to the acoustic-only version. This means that at least part of the observed separation was not due to the speech signal per se, but to features that correlated with the group labels.

The results are not necessarily invalidated by this fact. They just need to be interpreted correctly. A speech-based classifier that uses demographic features is different from a classifier that isolates COPD-specific vocal features. This study cannot distinguish between the two. It demonstrates that the former was useful for classification. It does not demonstrate how much of the result was actually disease-specific.

There is another point that needs to be discussed here. It is possible that other sources of variability besides age and gender are relevant. It is always challenging to distinguish between disease effects and cohort effects in a dataset like this. This is a general problem in speech-based health classification studies. It is a problem here as well.

C. No external test cohort

All of the key results were achieved via grouped cross-validation on one labeled data set. This was the correct approach for the project, and it was much better than a split that did not take speaker identity into account. Nevertheless, it is an internal validation. The models have not been validated

on an entirely independent data set that was recorded under different conditions.

The results showed that the process works better than chance for this data set. They do not demonstrate that the process would work as well for new speakers from a different data set, with a different microphone setup, or under a different protocol.

D. Controlled tasks only

The speech data for the project was controlled. This is a good thing for analysis because it reduces extraneous sources of data variation. However, it also makes the analysis easier. The downside is that it does not reflect the real world as well. Read speech, vowels derived from written passages, and sustained vowels are not the same as real-world speech.

A clinical application would not be done under controlled conditions. The real-world data would have noise, differences in microphone pickup, hesitations, interruptions, and differences in speaking styles. A process that does well on controlled data may not perform so well when all of the above are factored in.

E. Binary labels simplify a heterogeneous condition

The binary labels were used to represent the condition as either COPD or non-COPD. This made the problem simple. However, the binary labels were an over-simplification of the actual condition. COPD is not a single, uniform disease. There are many different factors for the disease, including severity, symptoms, breathing difficulties, and the general state of the patient.

This makes the results difficult to understand because, if the model is identifying a good signal, it is not clear whether the signal is related to the disease, breathing difficulties, voice stability, changes in the patient's pace, or a combination of factors. The same problem arises with the control side, as non-COPD is not necessarily the same as a more specific label of a healthy patient.

F. Dependence on segmentation and preprocessing quality

There are a number of preprocessing steps that the model depends on, including segmentation, alignment, feature extraction, and the combination of labels with the acoustic tables. This can lead to a number of errors, which is especially a problem for the passage-vowel branch because the segments are short, so a small change can have a large effect on the segment.

The results also showed that the raw inventories and the final labeled tables were not identical. There were some speakers who were represented in the segmentation inventories but were absent from the final labeled modeling tables. There were also some coverage mismatches for the sustained-vowel branch. All of these were addressed during preprocessing. However, it does indicate that the research did follow a rather complex filtering and preprocessing pathway.

G. Limited feature and model scope

The project focused on the specific set of eGeMAPS acoustic features and a small set of classical machine-learning approaches. This was a reasonable decision given the scope of this study.

The study did not demonstrate that these are the best possible set of features or that these are the best possible set of models for this specific problem. It demonstrates what was observed given this particular feature-set and model-set combination. It is possible that other approaches could be even better. As a result, this paper should not be interpreted as demonstrating that these are the final numbers for speech-based COPD classification on Danish datasets.

H. What these limitations mean for the study as a whole

Considering these issues together, we see that they limit the scope of this result. This study demonstrates that speech classification is possible, but it did not demonstrate strong generalization or disease-specific signal.

This is a useful result, but its scope is limited. This study is best seen as a validation step, not a final solution, and not a final demonstration of speech-based COPD detection ability.

VIII. Conclusion

The goal of this project was to determine whether there is enough acoustic information in standardized Danish speech to distinguish between people with COPD and those without. The answer is yes, to a moderate degree. The best phrase-based speech pipeline for grouped speaker evaluation had a speaker AUC of about 0.72, while the best passage-vowel speech pipeline had an AUC of about 0.67. This indicates that the speech does contain useful information for the task, although it is certainly far from solved.

Another important result is the comparison between the two main read-speech branches. The results indicate that the phrase-based approach is better than the passage-vowel approach on the main evaluation metric. This is likely due to the fact that the phrase-based approach preserves more timing information than isolated vowels. This does not mean that the isolated-vowel approach is bad for the task. Rather, it indicates that it is the weaker approach for this task.

The results must be interpreted in light of the small cohort, the controlled tasks, the internal validation design, and the likely demographic confounding.

The results of the project show that it is indeed possible to use a speaker-aware speech pipeline to detect above-chance structure related to COPD status in Danish speech. The results also show that the type of speech unit is important for the task. The results also provide a reasonable basis for further research with a larger number of participants, better control of confounders, and external validation.

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